

Case Study 1: Investigating Harassment Email Traffic With Wireshark

1. Case reference and student identification

Item	Detail
Full name	ABUBAKAR MUSTAPHA SADIQ
Registration number	2025/FWSD/11202
Course code	SBT-DF204
Case title	Investigating Harassment Email Traffic With Wireshark

2. Evidence acquisition and integrity (Question 1)

Item	Record
Original file name	nitroba.pcap
Source link	https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
File size	56,180,821 bytes (about 54 MB)
Download date/time	Fri 2 Oct 2026, 17:12:59 to 17:13:52 EDT (wget, Kali Linux VM)
SHA-256 (original)	2b77a9eafc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb [verify against your terminal]
Published checksum	Scenario page lists the same SHA-256 (and SHA-1 65656392412add15f93f8585197a8998aaeb50a1). Values match.
Working copy	~/SBTDF204_CaseStudy1/working/nitroba_working.pcap, made with cp; SHA-256 re-calculated and identical to the original; original set read-only (chmod 444).
Capture size	94,410 packets
Evidence refs	E01, E01a, E01b, E01c (see Appendix B)

3. Method

Filters and steps used, in order (tested against the capture):

- Opened the working copy only; set View > Time Display Format to UTC Date and Time of Day.
- `http.request.method == "GET" && http.host == "www.willselfdestruct.com"` (9 packets) to find the client and service.
- `ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"` (1 packet), then Follow TCP Stream (stream 1707).
- Inspected Ethernet II in packet 83601 for the MAC addresses; `eth.addr == 00:17:f2:e2:c0:ce` (73,246 packets) to trace the device.
- `eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"` (149 packets), then `http.cookie contains "gmailchat=jcoachj"` (100 packets) and its complement (49 packets, inspected).
- `frame.number in {82936, 83326, 83601, 83604, 83614, 84201}` to assemble the timeline.

4. Findings

Q1. File acquired and integrity

See section 3. Evidence: E01.

Q2. Client and service systems

Item	Value	Source
Client IP	192.168.15.4	Packet 82936, filter: GET to www.willselfdestruct.com; E02, E03
Service IP	69.25.94.22	Same packet (destination)
Time	2008-07-22 06:03:43.825871Z	Packet 82936, GET /secure/submit

Q3. Link between the client and the harassment message

Item	Observation
Packet / time	83601, 2008-07-22 06:04:24.311700Z, POST /secure/submit, TCP stream 1707 (E04, E05)
Form fields	to = lilytuckrige@yahoo.com; from = (empty); subject = you can't find us; message = and you can't hide from us. / Stop teaching. / Start running.; ttl = 30
Server reply	Packet 83604, 302 Moved Temporarily to /secure/success; confirmation page requested in packet 83614
Second email (scenario slides 10-12)	Same recipient, subject and message text; sender shown as undisclosed (E06, E06b, E06c)
Browser signature	User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)

Q4. Device that made the request

Item	Observation
Client MAC	00:17:f2:e2:c0:ce (Wireshark: Apple_e2:c0:ce), source of packet 83601 [verify by copying from Wireshark] (E08)
Next-hop MAC	00:1d:d9:2e:4f:60 (HonHaiPrecis_2e:4f:60), probably the gateway/router (inference)
Device activity	eth.addr == 00:17:f2:e2:c0:ce shows 73,246 of 94,410 packets (E09)

Q5. Identity evidence linked to the device

Item	Observation
Cookie value	gmailchat=jcoachj@gmail.com/475090, sent to mail.google.com (E10, E11)
Example packet	83326, 2008-07-22 06:04:05.546193Z, POST /mail/channel/bind
Frequency	100 of 149 cookie packets from this MAC carry this value; the other 49 contain no second account (analytics / ad cookies matched "@" by chance) (E12)

Q6. Roster comparison

Item	Observation
Roster	Chemistry 109 class list, scenario slide 14 (E07): 11 students
Comparison	Account name jcoachj contains "coach" and begins with "j"; the roster entry Johnny Coach is the only surname match. Other J-names (Jeremy Ledvkin, Jenny Kant) do not fit.
Gap	The trailing "j" is unexplained; account ownership is not stated in the scenario.

Q7. When the activity occurred

See section 6 (timeline). Wireshark's packet list was set to UTC.

Q8. Defensible conclusion

The packet capture shows that a device with IP address 192.168.15.4 and MAC address 00:17:f2:e2:c0:ce submitted a message to www.willselfdestruct.com at 06:04:24 UTC on 22 July 2008 (packet 83601). The recipient, subject and message text match the second email Ms Tuckrige reported. The same device repeatedly sent a Gmail cookie for jcoachj@gmail.com 100 packets, including about 19 seconds before the submission. The evidence therefore supports a link between this device and the Chem 109 student, but it does not prove who was typing.

I rate my confidence as moderate because reason in favour, for example the same device sent both the Gmail cookie and the harassment POST within seconds but against, for example nothing shows who was using the device.

6. Timeline (UTC, 22 July 2008)

UTC time	Packet	Direction	Event
06:03:43.825871	82936	192.168.15.4 to 69.25.94.22	GET /secure/submit (form page)
06:04:05.546193	83326	192.168.15.4 to 74.125.19.17	POST to Gmail; cookie shows jcoachj@gmail.com
06:04:24.311700	83601	192.168.15.4 to 69.25.94.22	POST /secure/submit (message submitted)
06:04:24.469169	83604	69.25.94.22 to 192.168.15.4	302 redirect to /secure/success
06:04:24.564165	83614	192.168.15.4 to 69.25.94.22	GET /secure/success (confirmation)
06:04:55.380722	84201	192.168.15.4 to 74.125.19.17	GET to Gmail; same session continues

Time-zone notes:

- Wireshark packet list: UTC (Z suffix). Capture File Properties showed the first packet as 2008-07-21 21:51:07 because the Kali VM displays EDT (UTC-4); this equals 01:51:07 UTC on 22 July.
- The Yahoo screenshot shows 11:04 PM, Mon 7/21/08. If the mailbox is at UTC-7 (the first email's header shows -0700), this equals 06:04 UTC on 22 July. This is an assumption, not directly shown.
- The willselfdestruct server's Date header in stream 1707 reads Tue, 22 Jul 2008 07:24:45 GMT, about 1 hour 20 minutes later than the capture timestamp (06:04:24Z); the clocks disagree and no explanation is offered here.

Appendix A. Evidence log

ID	Packet / item	Finding	Why it matters	Screenshot
E01	nitroba.pcap	SHA-256 2b77a9ea...553ec2fb matches published value	Confirms file integrity	E01_published_hash.png
E02	Packets 82936-83654	9 HTTP GETs to www.willselfdestruct.com from 192.168.15.4	Identifies web service activity	E02_GET_filter_results.png
E03	82936	Client 192.168.15.4, service 69.25.94.22, 06:03:43.825871Z	Identifies the two endpoints	E03_client_and_service_IP.png
E04	83601	POST /secure/submit with to/subject/message fields	Content matches reported message	E04_POST_request.png
E05	Stream 1707	Full request and 302 reply	Shows server accepted submission	E05_TCP_stream.png
E06	Slides 10-12	Second email: same recipient, subject, text	Correlates POST to complaint	E06_second_email.png, E06b, E06c
E07	Slide 14	Chem 109 roster (11 students)	Basis for roster comparison	E07_roster.png

ID	Packet / item	Finding	Why it matters	Screenshot
E08	83601 Ethernet II	Client MAC 00:17:f2:e2:c0:ce; gateway 00:1d:d9:2e:4f:60	Identifies the device	E08_client_MAC.png
E09	eth.addr filter	73,246 packets involve the client MAC	Shows device activity	E09_MAC_filter.png
E10	149 packets	Cookies containing "@" from the client MAC	Starting point for identity evidence	E10_cookie_filter.png
E11	83326 (+99 more)	gmailchat=jcoachj@gmail.com/475090 in 100 packets	Links device to a Gmail account	E11_cookie_identity.png
E12	49 packets	No second account; analytics/ad cookies	Rules out other identity values in this set	E12_other_cookie_packets.png
E13	6 packets	Timeline packets 82936 to 84201	Chronology	E13_timeline_packets.png

Appendix B. Labelled screenshots

Session tokens in cookie headers and URLs have been blacked out. The filter bar, packet list, packet detail and time are visible where possible.

E01a: Download of nitroba.pcap (size 56,180,821 bytes; date 2 Oct 2026)

```

kali@kali: ~/SBTDF204_CaseStudy1/original
Session Actions Edit View Help

(kali@kali)~/SBT-DF203-Lab9
$ mkdir -p ~/SBTDF204_CaseStudy1/original ~/SBTDF204_CaseStudy1/working

(kali@kali)~/SBT-DF203-Lab9
$ ls
evidence exported reports screenshots working

(kali@kali)~/SBT-DF203-Lab9
$ cd ~/SBTDF204_CaseStudy1

(kali@kali)~/SBTDF204_CaseStudy1
$ ls
original working

(kali@kali)~/SBTDF204_CaseStudy1
$ cd ~/SBTDF204_CaseStudy1/original

(kali@kali)~/SBTDF204_CaseStudy1/original
$ wget https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
--2026-10-02 17:12:59-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com) ... 52.92.203.121, 52.92.200.161, 52.92.179.225, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.203.121|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap'

nitroba.pcap      ~/SBT-DF203-Lab9      100%[=====] 53.58M  1.49MB/s   in 48s

2026-10-02 17:13:52 (1.11 MB/s) - 'nitroba.pcap' saved [56180821/56180821]

(kali@kali)~/SBTDF204_CaseStudy1/original
$ ls -l
total 54868
-rw-rw-r-- 1 kali kali 56180821 Nov 21 2020 nitroba.pcap

(kali@kali)~/SBTDF204_CaseStudy1/original
$ date
Fri Oct 2 05:14:21 PM EDT 2026

(kali@kali)~/SBTDF204_CaseStudy1/original
$

```

E01b: SHA-256 of the original, creation of the working copy, and identical SHA-256 of the copy

```
kali@kali: ~/SBTDF204_CaseStudy1/original
Session Actions Edit View Help

(kali@kali)~/SBTDF204_CaseStudy1/original
$ wget https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
--2026-10-02 17:12:59-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com) ... 52.92.203.121, 52.92.200.161, 52.92.179.225, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.203.121|:443 ... connected.
HTTP request sent, awaiting response ... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'nitroba.pcap'

nitroba.pcap 100%[=====] 53.58M 1.49MB/s in 48s

2026-10-02 17:13:52 (1.11 MB/s) - 'nitroba.pcap' saved [56180821/56180821]

(kali@kali)~/SBTDF204_CaseStudy1/original
$ ls -l SBTDF204_CaseStudy1/original
total 54868
-rw-rw-r-- 1 kali kali 56180821 Nov 21 2020 nitroba.pcap

(kali@kali)~/SBTDF204_CaseStudy1/original
$ date
Fri Oct 2 05:14:21 PM EDT 2026

(kali@kali)~/SBTDF204_CaseStudy1/original
$ sha256sum nitroba.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befd1a0b01f8c67553ec2fb nitroba.pcap

(kali@kali)~/SBTDF204_CaseStudy1/original
$ sha256sum nitroba.pcap > ../original_hash.txt

(kali@kali)~/SBTDF204_CaseStudy1/original
$ cp ~/SBTDF204_CaseStudy1/original/nitroba.pcap ~/SBTDF204_CaseStudy1/working/nitroba_working.pcap

(kali@kali)~/SBTDF204_CaseStudy1/original
$ sha256sum ~/SBTDF204_CaseStudy1/working/nitroba_working.pcap
2b77a9eafc1d6af163d1ba793c96dbccac04e6befd1a0b01f8c67553ec2fb /home/kali/SBTDF204_CaseStudy1/working/nitroba_working.pcap

(kali@kali)~/SBTDF204_CaseStudy1/original
$ chmod 444 ~/SBTDF204_CaseStudy1/original/nitroba.pcap

(kali@kali)~/SBTDF204_CaseStudy1/original
$
```

E01c: Wireshark Capture File Properties (SHA-256 of the working copy; times shown in EDT)

Wireshark - Capture File Properties - nitroba_working.pcap

File

Name: /home/kali/SBTDF204_CaseStudy1/working/nitroba_working.pcap
Length: 56 MB
Hash: 2b77a9eafc1d6af163d1ba793c96dbccac04e6befd1a0b01f8c67553ec2fb
(SHA256):
Hash (SHA1): 65656392412add15f93f8585197a8998aaeb50a1
Format: Wireshark/tcpdump/... - pcap
Encapsulation: Ethernet
Snapshot length: 4096

Time

First packet: 2008-07-21 21:51:07
Last packet: 2008-07-22 02:13:47
Elapsed: 04:22:39

Capture

Hardware: Unknown
OS: Unknown
Application: Unknown

Interfaces

Interface	Interface Description	Dropped packets	Capture filter	Link type	Packet size limit (snaplen)
Unknown	Unknown	Unknown	Unknown	Ethernet	4096 bytes

Statistics

Measurement	Captured	Displayed	Marked
Refresh	Edit Comments	Close	Copy To Clipboard

Warning: program compiled against libxml 213 using older 214

E01: Hash values published on the scenario page (SHA-256 matches)

1234

Restore Session816musty/SBT-DF203-La2008 Nitroba University Har...

digitalcorpora.org/corpora/scenarios/nitroba-university-harassment-scenario/

OffSecKali LinuxKali ToolsKali DocsKali ForumsKali NetHunterExploit-DBGoogle Hacking DB

Digital Corpora

Sponsored by the AWS Open Data Sponsorship Program

About DigitalCorporaHomeCorpora

Chem 109 roster. Your job is to determine if one of the students in the class was responsible for the harassing email and to provide clear, conclusive evidence to support your conclusion.

The teaching materials for this scenario includes:

- pcap file – about 60MB
- slides introducing the problem [PDF] [PPT] [Keynote]
- Password Protected Teacher's solution (password information)
- Expanded Nitroba Harassment Lab Scenario (Encrypted Solutions) (Thanks to Ajoy Ghosh at the UNSW Canberra for this work!)

Hash values for nitroba.pcap:

Algorithm	Value
MD5	9981827f11968773ff815e39f5458ec8
SHA1	65656392412add15f93f8585197a8998aeb50a1
SHA256	2b77a9aefc1d6af163d1ba793c96dbccab04e6befdf1a0b01f8c67553ec2fb

(Note: Because packet capture files contain timestamps for each packet, this scenario needs to have a

Recent Comments

[Simon Garfinkel on 2009 M57-Jean](#)

[Paramanandam on Govdocs1](#)

[Desmond on 2012 National Gallery DC Attack](#)

[Simon Garfinkel on 2012 National Gallery DC Attack](#)

[Desmond on 2012 National Gallery DC Attack](#)

E02: Filter http.request.method == GET && http.host == www.willselfdestruct.com (9 packets; first is 82936)

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.request.method == "GET" && http.host == "www.willselfdestruct.com"

No.	Time	Source	Destination	Protocol	Length	Info
82936	2008-07-22 06:03:43.825871Z	192.168.15.4	69.25.94.22	HTTP	596	GET /secure/submit HTTP/1.1
82985	2008-07-22 06:03:44.125949Z	192.168.15.4	69.25.94.22	HTTP	355	GET /images/spacer.gif HTTP/1.1
83025	2008-07-22 06:03:44.334254Z	192.168.15.4	69.25.94.22	HTTP	356	GET /images/sm-logo.gif HTTP/1.1
83037	2008-07-22 06:03:44.374583Z	192.168.15.4	69.25.94.22	HTTP	361	GET /images/warning-home.gif HTTP/1.1
83072	2008-07-22 06:03:44.530352Z	192.168.15.4	69.25.94.22	HTTP	356	GET /images/body-bk.gif HTTP/1.1
83087	2008-07-22 06:03:44.599079Z	192.168.15.4	69.25.94.22	HTTP	358	GET /images/btn-send.gif HTTP/1.1
83162	2008-07-22 06:03:44.852265Z	192.168.15.4	69.25.94.22	HTTP	361	GET /images/bridge_small.gif HTTP/1.1
83614	2008-07-22 06:04:24.564165Z	192.168.15.4	69.25.94.22	HTTP	461	GET /secure/success HTTP/1.1
83654	2008-07-22 06:04:24.902157Z	192.168.15.4	69.25.94.22	HTTP	360	GET /images/bk-message.gif HTTP/1.1

Frame 82936: Packet, 596 bytes on wire (4768 bits), 596 bytes captured on interface 0

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPre

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22

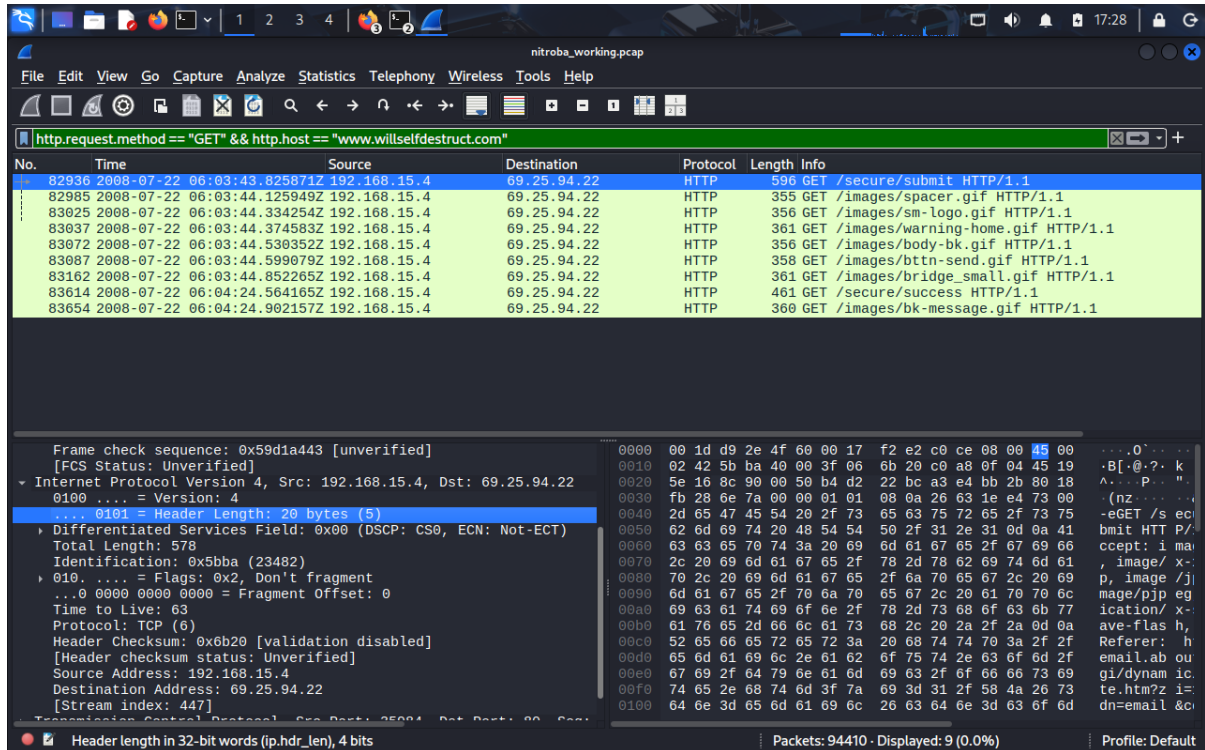
Transmission Control Protocol, Src Port: 35984, Dst Port: 80, Seq: 1

Hypertext Transfer Protocol

0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00 ...0...
0010 02 42 5b ba 40 00 3f 06 0b 20 c0 a8 0f 04 45 19 ...B[@? k
0020 5e 16 8c 90 00 50 b4 d2 22 bc a3 e4 bb 2b 80 18 ...A...P...
0030 fb 28 6e 7a 00 00 01 01 08 0a 26 03 1e e4 73 00 ... (nz ...
0040 2d 65 47 45 54 20 2f 73 65 63 75 72 65 2f 73 75 ...-eGET /s ec
0050 62 6d 69 74 20 48 54 54 50 2f 31 2e 31 0d 0a 41 ...bmit HTT P/
0060 63 63 65 70 74 3a 20 69 6d 61 67 65 2f 67 69 66 ...ccept: i ma
0070 2c 20 69 6d 61 67 65 2f 78 2d 78 62 69 74 6d 61 ...p, image/ x-
0080 70 2c 20 69 6d 61 67 65 2f 6a 70 65 67 2c 20 69 ...p, image/ j
0090 6d 61 67 65 2f 70 6a 70 65 67 2c 20 61 70 70 6c ...mage/ pjp eg
00a0 69 63 61 74 69 6f 6e 2f 78 2d 73 68 6f 63 6b 77 ...ication/ x-
00b0 61 76 65 2d 66 6c 61 73 68 2c 20 2a 2f 2a 0d 0a ...ave-flas h,
00c0 52 65 66 65 72 65 72 3a 20 68 74 74 70 3a 2f 2f ...Referer: h
00d0 65 6d 61 69 6c 2e 61 62 6f 75 74 2e 63 6f 6d 2f ...email.ab ou
00e0 67 69 2f 64 79 6e 61 6d 69 63 2f 6f 66 66 73 69 ...gi/dynam ic
00f0 74 65 2e 68 74 6d 3f 7a 69 3d 31 2f 58 4a 26 73 ...te.htm?z i=
0100 64 6e 3d 65 6d 61 69 6c 26 63 64 6e 3d 63 6f 6d ...dn=email &c

nitroba_working.pcapPackets: 94410 - Displayed: 9 (0.0%)Profile: Default

E03: Packet 82936, IPv4 section: client 192.168.15.4, service 69.25.94.22



The screenshot shows the Wireshark interface with packet 82936 selected. The packet list pane shows a series of HTTP GET requests from 192.168.15.4 to 69.25.94.22. The packet details pane shows the structure of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header.

Filter: `http.request.method == "GET" && http.host == "www.willselfdestruct.com"`

No.	Time	Source	Destination	Protocol	Length	Info
82936	2008-07-22 06:03:43.825871Z	192.168.15.4	69.25.94.22	HTTP	596	GET /secure/submit HTTP/1.1
82985	2008-07-22 06:03:44.125949Z	192.168.15.4	69.25.94.22	HTTP	355	GET /images/spacer.gif HTTP/1.1
83025	2008-07-22 06:03:44.334254Z	192.168.15.4	69.25.94.22	HTTP	356	GET /images/sm-logo.gif HTTP/1.1
83037	2008-07-22 06:03:44.374583Z	192.168.15.4	69.25.94.22	HTTP	361	GET /images/warning-home.gif HTTP/1.1
83072	2008-07-22 06:03:44.530352Z	192.168.15.4	69.25.94.22	HTTP	356	GET /images/body-bk.gif HTTP/1.1
83087	2008-07-22 06:03:44.599079Z	192.168.15.4	69.25.94.22	HTTP	358	GET /images/btn-send.gif HTTP/1.1
83162	2008-07-22 06:03:44.852265Z	192.168.15.4	69.25.94.22	HTTP	361	GET /images/bridge_small.gif HTTP/1.1
83614	2008-07-22 06:04:24.564165Z	192.168.15.4	69.25.94.22	HTTP	461	GET /secure/success HTTP/1.1
83654	2008-07-22 06:04:24.902157Z	192.168.15.4	69.25.94.22	HTTP	360	GET /images/bk-message.gif HTTP/1.1

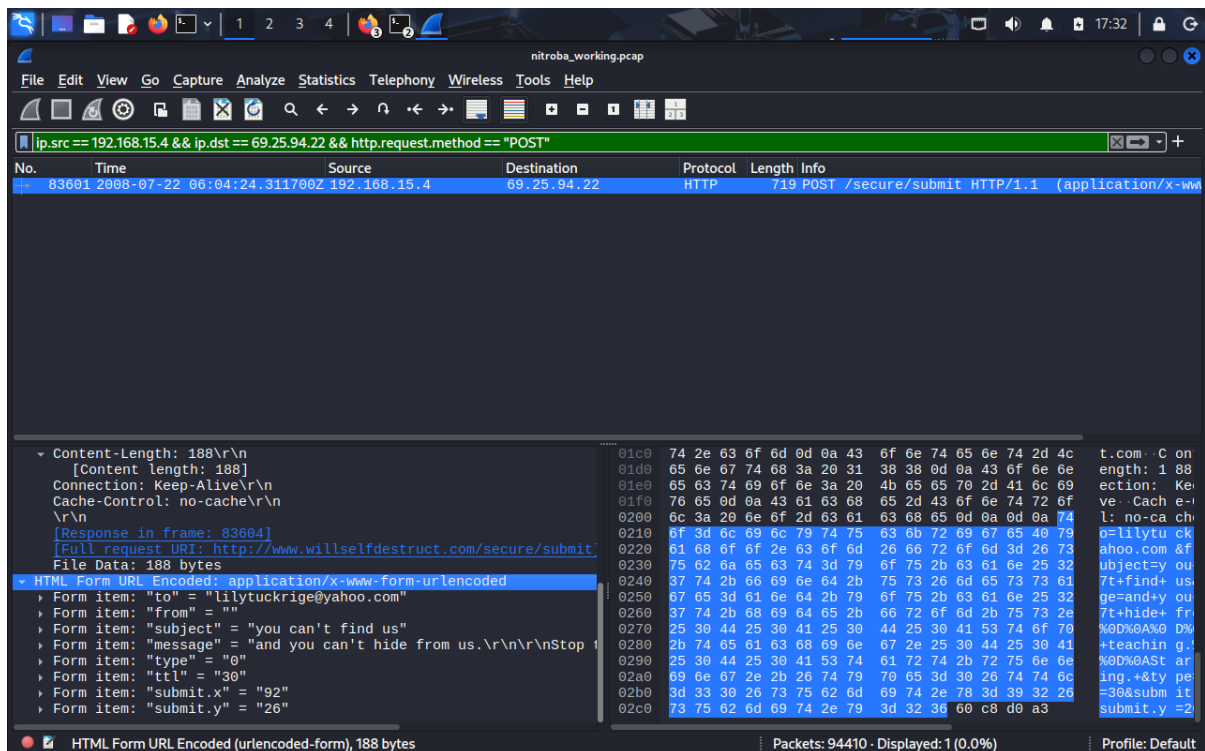
Frame check sequence: 0x59d1a443 [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 578
Identification: 0x5bba (23482)
010. = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 63
Protocol: TCP (6)
Header Checksum: 0x6b20 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.15.4
Destination Address: 69.25.94.22
[Stream index: 447]

Header length in 32-bit words (ip.hdr_len), 4 bits

Packets: 94410 - Displayed: 9 (0.0%)

Profile: Default

E04: Packet 83601, POST with HTML form fields (to, subject, message)



The screenshot shows the Wireshark interface with packet 83601 selected. The packet list pane shows a single HTTP POST request from 192.168.15.4 to 69.25.94.22. The packet details pane shows the structure of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Hypertext Transfer Protocol header.

Filter: `ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"`

No.	Time	Source	Destination	Protocol	Length	Info
83601	2008-07-22 06:04:24.311708Z	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)

Content-Length: 188\r\n
[Content length: 188]
Connection: Keep-Alive\r\n
Cache-Control: no-cache\r\n
\r\n
[Response in frame: 83604]
Full request URI: `http://www.willselfdestruct.com/secure/submit`
File Data: 188 bytes

HTML Form URL Encoded: `application/x-www-form-urlencoded`

- Form item: "to" = "lilytuckrige@yahoo.com"
- Form item: "from" = ""
- Form item: "subject" = "you can't find us"
- Form item: "message" = "and you can't hide from us.\r\n\r\nStop t"
- Form item: "type" = "g"
- Form item: "ttl" = "30"
- Form item: "submit.x" = "92"
- Form item: "submit.y" = "26"

HTML Form URL Encoded (urlencoded-form), 188 bytes

Packets: 94410 - Displayed: 1 (0.0%)

Profile: Default

E05: Follow TCP Stream, stream 1707 (POST request and 302 reply)

Wireshark - Follow TCP Stream (tcp.stream eq 1707) - nitroba_working.pcap

POST /secure/submit HTTP/1.1
Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*
Referer: http://www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Content-Length: 188
Connection: Keep-Alive
Cache-Control: no-cache

to=lilytuckrige@yahoo.com&from=&subject=you+can%27t+find+us&message=and+you+can%27t+hide+fr
om+us.%0D%0A%0D%0AStop+teaching.%0D%0A%0D%0Astart+running.+&type=0&ttl=30&submit.x=92&submi
t.y=26

HTTP/1.1 302 Moved Temporarily
Date: Tue, 22 Jul 2008 07:24:45 GMT
Server: Apache
Set-Cookie: [REDACTED]05A; Path=/
Location: http://www.willselfdestruct.com/secure/success
Content-Length: 0
Connection: close
Content-Type: text/plain; charset=UTF-8

Client pkt, 1 server pkt, 1 turn.
Entire conversation (938 bytes) Show as ASCII No delta times Stream 1707

Find: Case sensitive Find Next

Filter Out This Stream Print Save as... Back Close Help

Profile: Default

E06: Scenario slide 10, second email in the Yahoo inbox (Mon 7/21/08, 11:04 PM)

The guy attacked!

NITROBA State University

Window Help (1 unread) Yahoo! Mail, lilytuckrige

http://us.mg2.mail.yahoo.com/dc/launch?rand=euh4ui3e8u9oh

lilytuckrige Available | Yahoo! | News | Search the Web...

Home Inbox 2 messages Mobile | Options | Help

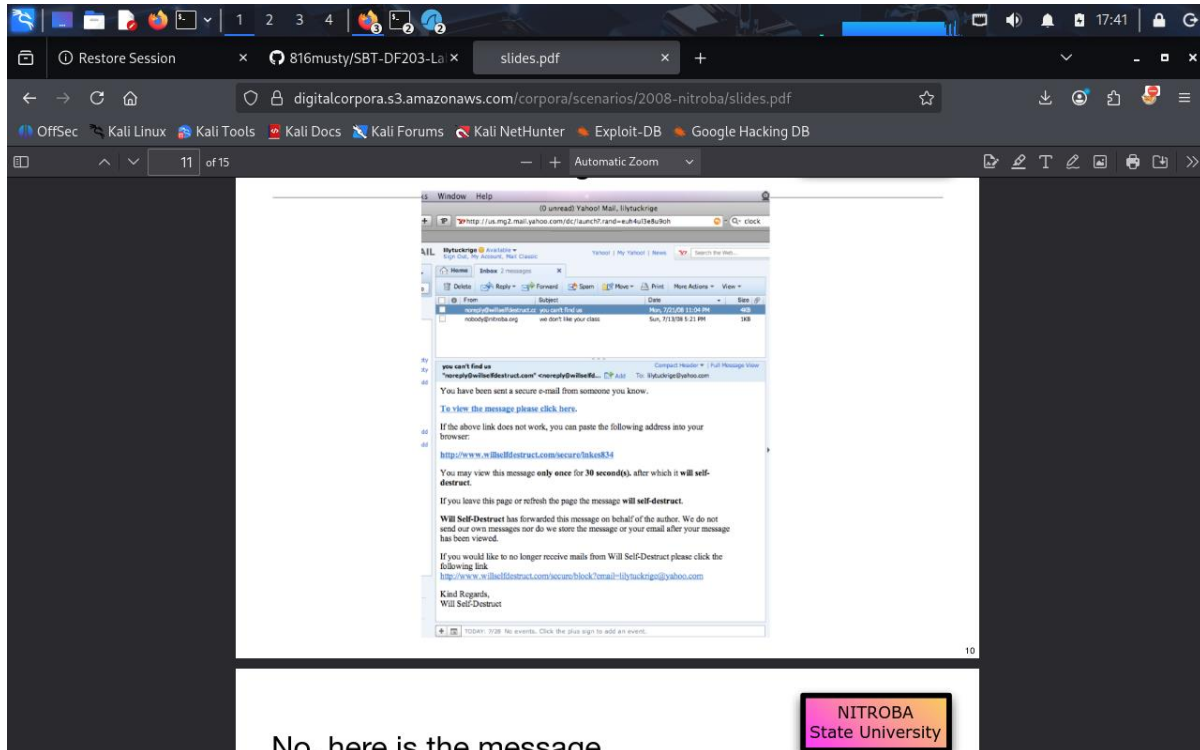
	From	Subject	Date	Size
☐	noreply@willselfdestruct	you can't find us	Mon, 7/21/08 11:04 PM	4KB
☐	nobody@nitroba.org	we don't like your class	Sun, 7/13/08 5:21 PM	1KB

Got your eye on one of those messages up there?
To view your message down here in this handy Reading Pane, just click on it.

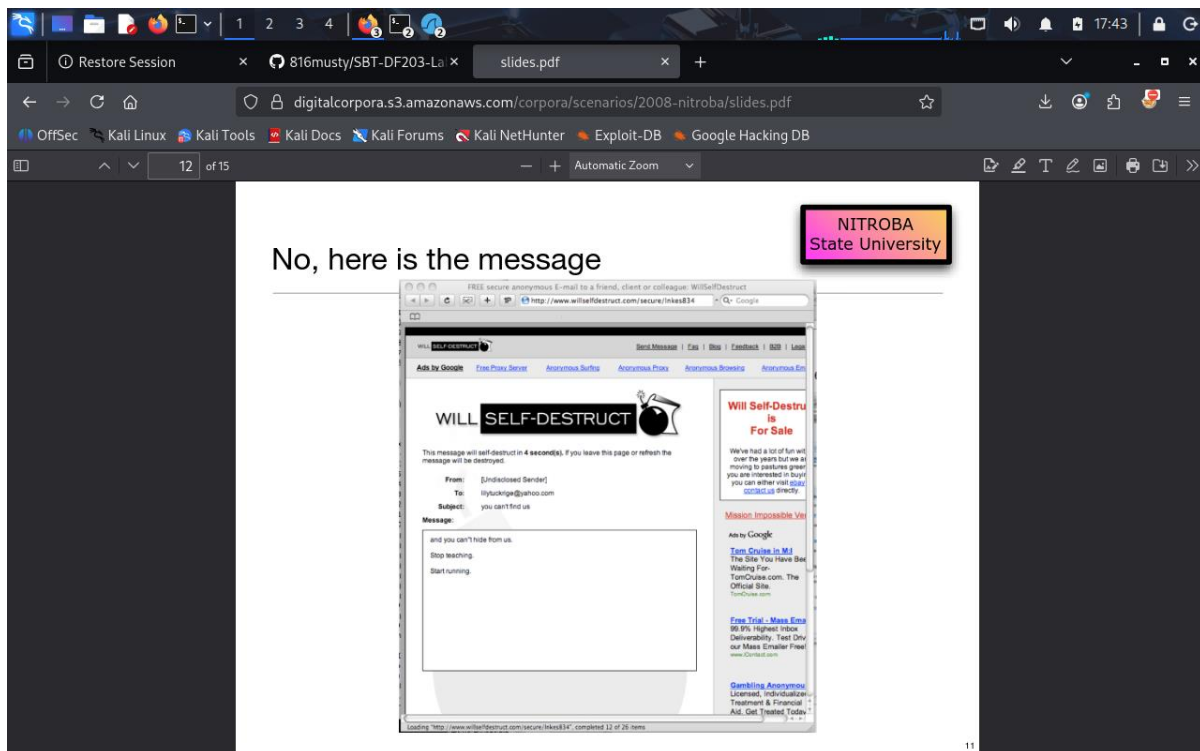
TODAY: 7/28. No events. Click the plus sign to add an event.

yahoo.com/dc/launch?rand=euh4ui3e8u9oh, completed 58 of 60 items (1 error)

E06b: Scenario slide 11, willselfdestruct notification email



E06c: Scenario slide 12, message shown by the service



E07: Scenario slide 14, Chemistry 109 class list

So who did it?

Chemistry 109 class list:

Teacher: Lily Tuckrige

Students:

- Amy Smith
- Burt Greedom
- Tuck Gorge
- Ava Book
- Johnny Coach
- Jeremy Ledvkin
- Nancy Colburne
- Tamara Perkins
- Esther Pringle
- Asar Misrad
- Jenny Kant

NITROBA State University

E08: Packet 83601, Ethernet II: source and destination MAC addresses

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.src == 192.168.15.4 && ip.dst == 69.25.94.22 && http.request.method == "POST"

No.	Time	Source	Destination	Protocol	Length	Info
83601	2008-07-22 06:04:24.311700Z	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)

Frame 83601: Packet, 719 bytes on wire (5752 bits), 719 bytes captured (5752 bits) on interface 0

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22

Transmission Control Protocol, Src Port: 36044, Dst Port: 80, Seq: 1000000000

Hypertext Transfer Protocol

HTML Form URL Encoded: application/x-www-form-urlencoded

Form item: "to" = "lilytuckrige@yahoo.com"

0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00 ...0? ...
0010 02 bd 02 ca 40 00 3f 06 c3 95 c0 a8 0f 04 45 19 ...@? ...
0020 5e 16 8c cc 00 50 fc 72 02 f9 75 b8 f4 26 80 18 ...P r ...
0030 fb 28 fa c7 00 00 01 01 08 0a 26 03 20 78 73 00 ...(. ...
0040 cb 95 50 4f 53 54 20 2f 73 65 63 75 72 65 2f 73 ...POST / se ...
0050 75 62 6d 69 74 20 48 54 54 59 2f 31 2e 31 0d 0a ...ubmit HT TP ...
0060 41 63 63 65 70 74 3a 20 69 6d 61 67 65 2f 67 69 ...Accept: im ...
0070 66 2c 20 69 6d 61 67 65 2f 78 2d 78 62 69 74 6d ...f, image /x ...
0080 61 70 2c 20 69 6d 61 67 65 2f 6a 70 65 67 2c 20 ...ap, image e ...
0090 69 6d 61 67 65 2f 70 6a 70 65 67 2c 20 61 70 70 ...image/pj pe ...
00a0 6c 69 63 61 74 69 6f 6e 2f 78 2d 73 68 6f 63 6b ...lication /x ...
00b0 77 61 76 65 2d 66 6c 61 73 68 2c 20 2a 2f 2a 0d ...wave-fla sh ...
00c0 0a 52 65 66 65 72 65 72 3a 29 68 74 74 70 3a 2f ...Referer : l ...
00d0 2f 77 77 77 2e 77 69 6c 6c 73 65 6c 66 64 65 73 .../www.wil ls ...
00e0 74 72 75 63 74 2e 63 6f 6d 2f 73 65 63 75 72 65 ...truct.co m/ ...
00f0 2f 73 75 62 6d 69 74 0d 0a 41 63 63 65 70 74 2d .../submit : A ...
0100 4c 61 6e 67 75 61 67 65 3a 29 65 6e 2d 75 73 0d ...Language : ...

nitroba_working.pcap

Packets: 94410 - Displayed: 1 (0.0%)

Profile: Default

E09: Filter eth.addr == 00:17:f2:e2:c0:ce (73,246 packets)

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

eth.addr == 00:17:f2:e2:c0:ce

No.	Time	Source	Destination	Protocol	Length	Info
83592	2008-07-22 06:04:19.053396Z	192.168.15.4	192.168.15.4	TCP	64	[TCP Retransmission] 80 → 35898 [FIN, ACK]
83593	2008-07-22 06:04:19.256189Z	192.168.15.4	208.185.127.40	TCP	64	35898 → 80 [ACK] Seq=299 Ack=387 Win=64768
83594	2008-07-22 06:04:22.546984Z	192.168.15.4	192.168.1.106	TCP	82	36042 → 3283 [SYN] Seq=0 Win=65535 Len=0 MSS=
83596	2008-07-22 06:04:23.483574Z	192.168.15.4	192.168.1.106	TCP	82	[TCP Retransmission] 36042 → 3283 [SYN] Seq=
83597	2008-07-22 06:04:24.213529Z	192.168.15.4	69.25.94.22	TCP	82	36044 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=
83598	2008-07-22 06:04:24.301080Z	69.25.94.22	192.168.15.4	TCP	78	80 → 36044 [SYN, ACK] Seq=0 Ack=1 Win=5792
83599	2008-07-22 06:04:24.305823Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=1 Ack=1 Win=64296 Len=
83600	2008-07-22 06:04:24.311077Z	192.168.15.4	69.25.94.22	TCP	70	35956 → 80 [FIN, ACK] Seq=365 Ack=1491980 Win=
83601	2008-07-22 06:04:24.311708Z	192.168.15.4	69.25.94.22	HTTP	719	POST /secure/submit HTTP/1.1 (application/javascript)
83602	2008-07-22 06:04:24.318712Z	69.25.94.22	192.168.15.4	TCP	70	80 → 35956 [FIN, ACK] Seq=1491980 Ack=366 Win=
83603	2008-07-22 06:04:24.406734Z	69.25.94.22	192.168.15.4	TCP	70	80 → 36044 [ACK] Seq=1 Ack=650 Win=6490 Len=
83604	2008-07-22 06:04:24.469169Z	69.25.94.22	192.168.15.4	HTTP	359	HTTP/1.1 302 Moved Temporarily
83605	2008-07-22 06:04:24.469263Z	69.25.94.22	192.168.15.4	TCP	70	80 → 36044 [FIN, ACK] Seq=290 Ack=650 Win=
83606	2008-07-22 06:04:24.471575Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=290 Win=64007
83607	2008-07-22 06:04:24.471964Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [ACK] Seq=650 Ack=291 Win=64007
83608	2008-07-22 06:04:24.472590Z	192.168.15.4	69.25.94.22	TCP	70	36044 → 80 [FIN, ACK] Seq=650 Ack=291 Win=

Frame 83601: Packet, 719 bytes on wire (5752 bits), 719 bytes captured (5752 bits) on interface 0
Ethernet II, Src: Apple_E2:C0:CE (00:17:f2:e2:c0:ce), Dst: HonHaiPrecision_2E:4F:60 (00:1d:d9:2e:4f:60)
Destination: HonHaiPrecision_2E:4F:60 (00:1d:d9:2e:4f:60)
Source: Apple_E2:C0:CE (00:17:f2:e2:c0:ce)
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x0c8d0a3 [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
Transmission Control Protocol, Src Port: 36044, Dst Port: 80, Seq: 290
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "to" = "lilytuckrige@yahoo.com"

Key (urlencoded-form.key), 2 bytes

Packets: 94410 · Displayed: 73246 (77.6%)

Profile: Default

E10: Filter eth.src == client MAC && http.cookie contains "@" (149 packets)

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"

No.	Time	Source	Destination	Protocol	Length	Info
79715	2008-07-22 06:01:16.853458Z	192.168.15.4	74.125.19.104	HTTP	1384	GET /calendar/render?utm_c
79732	2008-07-22 06:01:17.225783Z	192.168.15.4	74.125.19.17	HTTP	131	POST /mail/channel/bind?at=
79749	2008-07-22 06:01:17.425424Z	192.168.15.4	74.125.19.104	HTTP	151	POST /calendar/calddetails
79750	2008-07-22 06:01:17.425760Z	192.168.15.4	74.125.19.104	HTTP	336	POST /calendar/load HTTP/1
80225	2008-07-22 06:02:05.562559Z	192.168.15.4	74.125.19.17	HTTP	120	POST /mail/channel/bind?at=
80824	2008-07-22 06:03:02.386299Z	192.168.15.4	74.125.19.17	HTTP	1405	GET /mail/?ui=1&ik=a0610ea
80842	2008-07-22 06:03:05.557263Z	192.168.15.4	74.125.19.17	HTTP	121	POST /mail/channel/bind?at=
83217	2008-07-22 06:03:48.787555Z	192.168.15.4	76.13.208.11	HTTP	826	GET /st?ad_type=iframe&ad=
83326	2008-07-22 06:04:05.546193Z	192.168.15.4	74.125.19.17	HTTP	121	POST /mail/channel/bind?at=
84139	2008-07-22 06:04:48.915042Z	192.168.15.4	76.13.208.11	HTTP	846	GET /st?ad_type=iframe&ad=
84201	2008-07-22 06:04:55.380722Z	192.168.15.4	74.125.19.17	HTTP	1366	GET /mail/channel/bind?at=
84598	2008-07-22 06:05:08.525100Z	192.168.15.4	76.13.208.11	HTTP	861	GET /st?ad_type=iframe&ad=
86214	2008-07-22 06:06:31.130705Z	192.168.15.4	76.13.212.11	HTTP	956	GET /st?ad_type=iframe&ad=
86642	2008-07-22 06:07:30.143149Z	192.168.15.4	76.13.212.11	HTTP	957	GET /st?ad_type=iframe&ad=
87045	2008-07-22 06:07:51.016293Z	192.168.15.4	76.13.212.11	HTTP	957	GET /st?ad_type=iframe&ad=
87157	2008-07-22 06:07:51.699318Z	192.168.15.4	76.13.212.11	HTTP	730	GET /pixel?id=94635&t=2 H

Frame 83326: Packet, 121 bytes on wire (968 bits), 121 bytes captured (968 bits) on interface 0
Ethernet II, Src: Apple_E2:C0:CE (00:17:f2:e2:c0:ce), Dst: HonHaiPrecision_2E:4F:60 (00:1d:d9:2e:4f:60)
Destination: HonHaiPrecision_2E:4F:60 (00:1d:d9:2e:4f:60)
Source: Apple_E2:C0:CE (00:17:f2:e2:c0:ce)
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x122f1a01 [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 74.125.19.17
Transmission Control Protocol, Src Port: 35804, Dst Port: 80, Seq: 1443
[2 Reassembled TCP Segments (1443 bytes): #83324(1392), #83326(51)]
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "to" = "lilytuckrige@yahoo.com"

Packet (121 bytes) Reassembled TCP (1443 bytes)

nitroba_working.pcap

Packets: 94410 · Displayed: 149 (0.2%)

Profile: Default

E11: Packet 83326, cookie pair gmailchat=jcoachj@gmail.com/475090 (other tokens redacted)

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The packet list pane shows a series of packets, with packet 83326 selected, which is an HTTP GET request to /calendar/render?utm_g. The packet details pane shows the structure of the selected packet, including the request line, headers (Cache-Control, Cookie, User-Agent), and the body. The packet bytes pane shows the raw data of the request, including the request line, headers, and body. The status bar at the bottom indicates that the packet is a GET request (121 bytes) and that the reassembled TCP segment is 1443 bytes.

E11b: Filter http.cookie contains gmailchat=jcoachj (100 packets)

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The main packet list pane shows a list of captured packets, with packet 121 (HTTP GET) selected. The packet details pane on the right shows the structure of the selected packet, including the Cookie field which contains the value 'gmailchat=jcoachj@gmail.com/475090'. The packet bytes pane at the bottom shows the raw data of the packet.

E12: The other 49 packets (analytics and ad cookies, no second account)

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@" && !(http.cookie contains "gmailchat=jcoachj")

No.	Time	Source	Destination	Protocol	Length	Cookie pair
77547	2008-07-22 06:00:44.666574Z	192.168.15.4	74.125.19.104	HTTP	81	_utma=173272373.890237978.1216706402
77710	2008-07-22 06:00:45.360749Z	192.168.15.4	74.125.19.104	HTTP	151	_utma=173272373.890237978.1216706402
77711	2008-07-22 06:00:45.361069Z	192.168.15.4	74.125.19.104	HTTP	336	_utma=173272373.890237978.1216706402
77727	2008-07-22 06:00:50.381580Z	192.168.15.4	74.125.19.104	HTTP	84	_utma=173272373.890237978.1216706402
79670	2008-07-22 06:01:15.423593Z	192.168.15.4	74.125.19.104	HTTP	103	_utma=173272373.890237978.1216706402
79715	2008-07-22 06:01:16.853458Z	192.168.15.4	74.125.19.104	HTTP	1384	_utma=173272373.890237978.1216706402
79749	2008-07-22 06:01:17.425424Z	192.168.15.4	74.125.19.104	HTTP	151	_utma=173272373.890237978.1216706402
79750	2008-07-22 06:01:17.425760Z	192.168.15.4	74.125.19.104	HTTP	336	_utma=173272373.890237978.1216706402
83217	2008-07-22 06:03:48.787555Z	192.168.15.4	76.13.208.11	HTTP	826	testbounce="testing",bh="b!!!!1!!!"(h
84139	2008-07-22 06:04:48.915042Z	192.168.15.4	76.13.208.11	HTTP	846	testbounce="testing",bh="b!!!!1!!!"(h
84598	2008-07-22 06:05:08.525100Z	192.168.15.4	76.13.208.11	HTTP	861	testbounce="testing",bh="b!!!!1!!!"(h
86214	2008-07-22 06:06:31.130705Z	192.168.15.4	76.13.212.11	HTTP	956	testbounce="testing",bh="b!!!!1!!!"(h
86642	2008-07-22 06:07:30.143149Z	192.168.15.4	76.13.212.11	HTTP	957	testbounce="testing",bh="b!!!!1!!!"(h
87045	2008-07-22 06:07:51.016293Z	192.168.15.4	76.13.212.11	HTTP	957	testbounce="testing",bh="b!!!!1!!!"(h
87157	2008-07-22 06:07:51.699318Z	192.168.15.4	76.13.212.11	HTTP	730	testbounce="testing",bh="b!!!!1!!!"(h
87171	2008-07-22 06:07:51.766274Z	192.168.15.4	76.13.212.11	HTTP	825	testbounce="testing",bh="b!!!!1!!!"(h

Frame 83217: Packet, 826 bytes on wire (6608 bits), 826 bytes captured (6608 bits) on interface 0

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)

Type: IPv4 (0x0800)

Stream index: 19

Frame check sequence: 0x805ad097 [unverified]

[FCS Status: Unverified]

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 76.13.208.11

Transmission Control Protocol, Src Port: 36014, Dst Port: 80, Seq: 2152, Win: 0, Len: 0

[2 Reassembled TCP Segments (2152 bytes): #83215(1396), #83217(756)]

Hypertext Transfer Protocol

GET /st7ad_type=iframe&ad_size=160x600&site=140060§ion=...

Packet (826 bytes) Reassembled TCP (2152 bytes)

nitroba_working.pcap Packets: 94410 - Displayed: 49 (0.1%) Profile: Default

E13: Timeline packets 82936, 83326, 83601, 83604, 83614, 84201 in UTC

nitroba_working.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

frame.number in {82936, 83326, 83601, 83604, 83614, 84201}

No.	Time	Source	Destination	Protocol	Length	Cookie pair
82936	2008-07-22 06:03:43.825871Z	192.168.15.4	69.25.94.22	HTTP	596	
83326	2008-07-22 06:04:05.546193Z	192.168.15.4	74.125.19.17	HTTP	121	
83601	2008-07-22 06:04:24.311700Z	192.168.15.4	69.25.94.22	HTTP	719	
83604	2008-07-22 06:04:24.469169Z	192.168.15.4	192.168.15.4	HTTP	359	
83614	2008-07-22 06:04:24.564165Z	192.168.15.4	69.25.94.22	HTTP	461	
84201	2008-07-22 06:04:55.380722Z	192.168.15.4	74.125.19.17	HTTP	1366	

Frame 82936: Packet, 596 bytes on wire (4768 bits), 596 bytes captured (4768 bits) on interface 0

Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)

Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)

Type: IPv4 (0x0800)

Stream index: 19

Frame check sequence: 0x59d1a443 [unverified]

[FCS Status: Unverified]

Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22

Transmission Control Protocol, Src Port: 35984, Dst Port: 80, Seq: 2152, Win: 0, Len: 0

Hypertext Transfer Protocol

GET /secure/submit HTTP/1.1

Request Method: GET

nitroba_working.pcap Packets: 94410 - Displayed: 6 (0.0%) Profile: Default